Def Let X^{2n} be a smooth mfld, an almost complex structure on X is $J \in End(TX)$ s.t. $J^2 = -Id$. Notice this implies that we can find an atlas for X with values in $U(n)$

Let us focus on $\dim X = 4$ now.

Def An almost complex structure J is compatible with the Riemannian metric g if J is an orthogonal map fibrewise. A 2-form ω is compatible with the Riemannian metric if ω is a self dual 2-form w.r.t. g and $\|\omega\| = \sqrt{2}$. A 2-form ω and an almost-complex structure J are compatible if $\omega(Jv_1, Jv_2) = \omega(v_1, v_2)$ and $\omega(v, Jv) > 0$ $\forall v \neq 0$ in TX.

A 2-form ω that is compatible with some g or J is automatically non-degenerate and respect the given orientation. In fact, let ω compatible with g, then $\omega \wedge \omega \overset{\text{self-dual}}{=} \omega \wedge * \omega = \langle \omega, \omega \rangle vol_X$ and the latter is always non-zero since it's a multiple of the volume form. Now let ω compatible with J, let $v \in T_pX$ s.t. $\omega(v, z) = 0$ $\forall z \in T_pX$ but let $z = Jv$, then $0 = \omega(v, z) = \omega(v, Jv) > 0$ absurd if $v \neq 0$.

Remark It's not hard to see that in $GL_4(\mathbb{R})$ we have

$$SO(4) \cap Sp(4) = SO(4) \cap GL_2\mathbb{C} = Sp(4) \cap GL_2\mathbb{C} = U(2)$$

Hence every time we fix two compatible structures from g, J and ω, the third one is automatically defined, (for example $\omega(x,y) = g(x, Jy)$

$g(x,y) = \omega(x, Jy)$

$J\tilde{x} = -*(\omega \wedge \tilde{x})$ where $\tilde{x}$ is given by $TX \overset{g}{\cong} T^*X$

Notice that we did not assume anything about the global behaviour of J, g or ω.

Prop Let M be a smooth manifold.

i) For every non-degenerate 2-form ω on M, the space of compatible almost complex structures is non empty and contractible.

ii) Let J be an almost complex structure on M. Then the space of non-degenerate 2-forms on M that are compatible with J is non-empty and contractible.

iii) Fix a Riemannian metric g on M. Then the map $\omega \mapsto J_{g,\omega}$ induces a homotopy equivalence from the space of non-deg 2-forms on M to the space of a.c.s. on M.

A complex structure on X^4 naturally induces an almost complex structure $J: TX \to TX$ by the formula $Jz = i \cdot z$

Def An almost complex structure is called *integrable* if it arises in this manner from a complex structure

Def A manifold X is called *Kähler* if it admits ω, J, g compatible, $d\omega = 0$ and J is integrable.

If J is not integrable, we say that X is *almost-Kähler*

Example: [The complex projective space] This is the space $\mathbb{CP}^n$ of complex lines in $\mathbb{C}^{n+1}$. Thus a point in $\mathbb{CP}^n$ is the equivalence class of a non-zero complex $(n+1)$-vector $[z] = [z_0 : \dots : z_n] = [\lambda z_0 : \dots : \lambda z_n]$ for $\lambda \in \mathbb{C} \setminus 0$.

Let $U_j \subset \mathbb{CP}^n$ denote the coordinate patch where $z_j \neq 0$ and define

$$\phi_j : U_j \to \mathbb{C}^n : [z_0 : \dots : z_n] \mapsto \left(\frac{z_0}{z_j}, \dots, \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, \dots, \frac{z_n}{z_j} \right)$$

It's a standard fact that the transition function $\phi_k \circ \phi_j^{-1}$ are holomorphic and hence they induce an (integrable) a.c.s. call it J. Now consider the 2-form

$$\rho_{FS} := \frac{i}{2(\sum_{\nu=0}^{n} \bar{z}_\nu z_\nu)^2} \sum_{k=0}^{n} \sum_{j \neq k} \left(\bar{z}_j z_k \, dz_k \wedge d\bar{z}_j - \bar{z}_j z_k \, dz_j \wedge d\bar{z}_k \right)$$

This is a real valued 2-form on $\mathbb{C}^{n+1} \setminus 0$, since it's straightforward to see that if we evaluate it on real vector fields we get a real value. In fact a little manipulation of the summands shows that the result of that double sum is going to be purely imaginary, hence by multiplying by i we get a real value.

The normalization factor $\frac{1}{\|z\|^2}$ ensures that the form descends to a form on $\mathbb{CP}^n$. Let us call it ω_{FS}.

In other words, let $\pi : \mathbb{C}^{n+1} \setminus 0 \to \mathbb{CP}^n$, we have $\rho_{FS} = \pi^* \omega_{FS}$. By working on a local chart, one can see it's a closed form, i.e. $\omega_{FS} = \frac{i}{2} \sum_j \partial \bar{\partial} \log\left(\frac{\|z\|^2}{|z_j|^2} \right)$, hence $d\omega_{FS} = (\partial + \bar{\partial}) \omega_{FS} = 0$, using the relations $\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0$.

Non-degeneracy comes from the fact that the restriction of ρ_{FS} to S^{2n+1} agrees with the std symplectic form $\omega_0 = \sum_k dx_k \wedge dy_k$.

Prop Any symplectic manifold (X,ω) admits a compatible almost-complex structure J and (hence) a Riemann metric g s.t. (X,ω,J,g) is almost-kähler. The space of compatible almost complex structure is contractible.

Proof The proof rests on the fact that the space of a.c. structures on $\mathbb{R}^4$ compatible with ω_{std} is non-empty and contractible. It follows that the compatible a.c.s. on (X,ω) are sections of a bundle with contractible fibers, so they form a non empty, contractible space.

Theorem A (closed) complex surface S is kähler iff the first Betti number is even

Remark A complex surface S (with the induced almost-complex structure) is called kähler if it admits a metric s.t. (S,ω,J,g) is kähler. Since a complex submanifold of a kähler manifold is kähler, and $\mathbb{CP}^n$ admits one (the Fubini-Study metric), all projective manifolds are kähler.

Not all kähler manifold are projective, but we have the following result

Theorem [Kodaira Embedding theorem] A compact complex manifold X is projective if and only if there is a kähler form ω on X whose class in $H^2(X;\mathbb{R})$ is in the image of the integral cohomology group $H^2(X;\mathbb{Z})$ $\square$

Example Consider the hypersurface of degree d in $\mathbb{CP}^3$

$$X_d := \{ [z_0 : \cdots : z_3] \in \mathbb{CP}^3 \mid z_0^d + z_1^d + z_2^d + z_3^d = 0 \}$$

This is a smooth manifold. The Lefschetz hyperplane theorem implies that the inclusion $X_d \hookrightarrow \mathbb{CP}^3$ induces iso on π_0 and π_1 and hence X_d is connected and simply connected.

To apply Lefschetz hyperplane thm one observe that $X_d \cong H \cap V$, where H is an hyperplane and $V \cong \mathbb{CP}^3$. The theorem tells us that in this setting we have the desired isos on homotopy groups.

This implies that the 2^{nd} homology is generated by $\pi_2(X_d)$. We claim the following

$$b_2 := \operatorname{rank} H_2(X_d;\mathbb{Z}) = d^3 - 4d^2 + 6d - 2$$

$$\operatorname{sgn}(Q_{X_d}) = \frac{1}{3}(4-d^2)\, d$$

The proof of the last equality relies on the famous formula by Hirzebruch

$$\mathrm{sgn}(Q_X) = \frac{1}{3}\langle c_1(TX)^2 - 2c_2(TX), [X] \rangle$$

It's interesting to distinguish the cases $d \leq 3$, $d = 4$, $d \geq 5$. The manifolds $X_1 \cong \mathbb{CP}^2$, $X_2 \cong S^2 \times S^2$, $X_3 \cong \mathbb{CP}^2 \#_6 \overline{\mathbb{CP}^2}$ are positive in the sense that the cohomology class $[\omega]$ is a positive multiple of the first chern class. (They are called monotone symplectic manifolds)

The manifold X_4 is a compact, connected and simply connected 4-dim Kähler manifold with $c_1(TX_4) = 0$. All 4-mflds with these property are diffeomorphic. They have $b_2 = 22$ and are called K3 surfaces.

The manifold X_d, for $d \geq 5$ are surfaces of general type (algebraic surface with Kodaira dimension 2, in some sense most algebraic surfaces are in this class)